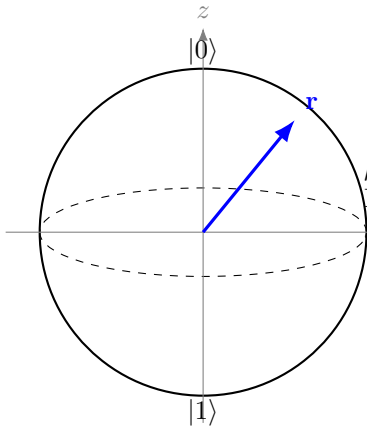




$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$$

Pauli matrices generate rotations
and represent spin observables.